

STUDY-UNI · FULL MARKING SCHEME

FE3020

Understanding Supply Chain Management

Institution	University College Cork
Examination session	Winter 2025
Paper number	Paper 1
Duration of paper	1.5 hours
Instructions to candidates	Answer two questions. All questions carry equal marks.
Questions covered in this scheme	Question 1, Question 2, Question 3 (all three, since candidates may choose any two)

This is an original marking scheme prepared for exam-preparation purposes. It is not an official UCC document and has no connection to the module's examiners or external examiner.

Question 1

50% + 50% = 100%

(a)**50%**

Using the three pillars of the ESG framework, critique the use of this framework to address sustainability issues at scopes 2 and 3. Support your answer with specific examples.

FULL MODEL ANSWER

The three pillars. ESG evaluates an organisation's sustainability performance across **Environmental** (resource use, emissions, waste, biodiversity), **Social** (labour conditions, community impact, health & safety, human rights across the value chain), and **Governance** (board oversight, disclosure, targets, and accountability structures). Applied to supply chains, the framework is meant to move sustainability from a peripheral concern (see the module's "true cost" and externalities discussion) into something structurally reported and managed.

SOURCING NOTE *The GHG Protocol's scope 1/2/3 emissions classification is standard carbon-accounting practice referenced across sustainability literature; it is not spelled out in the module slides beyond the ESG pillar titles, so the scope-specific critique below extends deliberately beyond what is explicit in the notes and should be marked as such.*

Scope 2 covers indirect emissions from purchased energy (electricity, heat, steam) — emissions a firm does not generate directly but pays for. **Scope 3** covers all other indirect emissions across the upstream and downstream value chain: supplier production, inbound and outbound logistics, and (for food businesses) ingredient sourcing and packaging. Scope 3 is typically the largest share of a food or consumer goods company's footprint by a wide margin, and the hardest to measure, because it depends on data from parties outside the firm's direct control.

Critique by pillar:

Environmental. This pillar is the natural home for scope 2/3 reporting, but a single aggregated "E" score can obscure whether scope 3 — the dominant and hardest-to-verify component for a food manufacturer sourcing from multiple ingredient suppliers — is even being measured, let alone reduced. A firm can score well on visible, controllable scope 1/2 items (e.g. switching a distribution centre to a renewable-backed electricity contract) while its much larger upstream footprint, sitting with small-scale speciality suppliers, goes largely unquantified.

Social. The social pillar has no direct mechanism for carbon scope boundaries at all — its natural link to scope 2/3 is indirect, via supplier due diligence (the module's reference to NGO scorecards, e.g. Oxfam, on labour and ethics in the same supplier tier where scope 3 emissions originate). The result is a structural blind spot: a supplier can score well on social criteria while the same relationship carries an unmeasured and unmanaged scope 3 carbon exposure, because the two are assessed on separate criteria that are never required to be read together.

Governance. This is where ESG has the most real leverage over scope 2/3, since governance criteria can mandate disclosure, target-setting, and supplier data-sharing (functionally similar to the CPFR-style information sharing discussed elsewhere in this module). The

critique here is that governance metrics typically assess whether a policy or target *exists*, not whether the underlying scope 3 data is accurate. Given the module's own point that sustainable supply chain redesign depends on coordination across fragmented, often small-scale suppliers, a governance framework can reward the presence of a target long before the measurement infrastructure behind it is trustworthy — creating a false sense of assurance.

Overall critique. ESG was originally designed as an investor risk-screening tool, not a supply-chain carbon-accounting system. Its three-pillar structure segments what, for scope 2/3 purposes, is a single integrated operational and data-quality problem spanning a multi-tier, often low-visibility supplier base. A framework built to summarise risk in a single score is a poor fit for a measurement problem whose central difficulty is upstream data granularity, not disclosure intent.

Examples. A retailer moving distribution centre electricity to a renewable power purchase agreement directly addresses scope 2, and is straightforward to verify. Addressing scope 3 is structurally harder: it requires supplier engagement programmes with the kind of small, speciality ingredient suppliers described later in this paper (CW Foods, Question 3(b)) — suppliers who may lack the resourcing to report emissions data at all, regardless of how strong the buying firm's governance pillar looks on paper.

Mark Allocation — Part (a) (worth 50% of the question)

Criterion	Marks	What earns the marks
Correctly explains the three ESG pillars	10%	Environmental, Social, Governance each clearly and accurately defined
Correctly distinguishes scope 2 vs scope 3	10%	Purchased-energy vs value-chain framing, AND notes scope 3 is typically dominant/hardest to measure
Critique of Environmental pillar	8%	Identifies aggregation problem — scope 3 can be masked within a single E score
Critique of Social pillar	8%	Identifies the disconnect between social due-diligence criteria and carbon scope boundaries
Critique of Governance pillar	8%	Identifies gap between target/policy existence and data-quality assurance
Specific, relevant examples	6%	At least one concrete scope 2 and one scope 3-relevant example, correctly applied
Total	50%	Sum of criteria above

(b)**50%**

John is a Demand Planner for a large global food manufacturing company that produces packaged snacks and beverages. The Research & Development Team Lead on a new snack flavour range has just informed him that the planned release date for market trials needs to be postponed by two months. This is due to ingredient licencing issues with an ingredient supplier which is not unusual. John is aware of the knock-on effect on market launch, production plans, etc. The business runs a standard Sales & Operations Planning (S&OP) Process as compared to more 'advanced' S&OP Process, as discussed in this module. How should John use the current S&OP process to address this issue? How could the S&OP Process currently in use be improved to better manage this type of issue in the future.

FULL MODEL ANSWER

Using the current process to address the issue now. The module's standard five-step S&OP cycle gives John a clear route: (1) gather sales/marketing data to build a baseline demand forecast (typically a 6–18 month horizon); (2) gather inventory and capacity information to build an initial supply plan against that baseline; (3) hold the cross-functional S&OP meeting — sales, marketing, operations, purchasing, logistics and finance — to agree the final operating plan; (4) distribute and implement the plan, primarily through operations and sales; (5) measure the results.

John should not wait for the next scheduled cycle: he should raise the two-month delay immediately so it is reflected at step 1 (the baseline forecast for this SKU range is revised to reflect the later release) and escalated to step 3, the cross-functional meeting, so operations, purchasing, and marketing can jointly re-sequence the launch. This matters because the module's New Product Introduction guidance is explicit that new SKUs must sit in the Master Production Schedule, that forecast accuracy is inherently lower for new than for mature products, and that inventory holding policy for new products should differ from mature-product policy for exactly this reason. Left unmanaged, the two-month slip risks premature production of packaging/ingredients tied to the original date, wasted capacity allocation, and a mismatched marketing launch — effectively a self-inflicted bullwhip event triggered by a delay that (per the question) "is not unusual."

SOURCING NOTE *The five-step process and NPI rules above are drawn directly from the module's S&OP content. The 'advanced S&OP' / Integrated Business Planning (IBP) framing below is standard supply chain practice used to answer the comparative half of the question, and is not itself named in the module slides — it should be credited on the strength of the reasoning, not as a recall of a specific slide term.*

Improving the process for the future. Three specific improvements follow directly from the gap between a standard, calendar-driven S&OP cycle and a more advanced one:

- **Build supplier/licensing risk into the S&OP data feed.** The module's own recommendation is to "build in review of new product launch dates and capacity allocation into the S&OP monthly cycle" — this should explicitly include known supplier-side risk factors (ingredient licensing status among them) rather than treating a licensing delay as an external shock discovered after the fact.

- **Move toward continuous/rolling planning rather than a fixed monthly cycle** for NPI-heavy product families, so a two-month schedule slip can be absorbed and re-planned without waiting for the next calendar S&OP meeting — reducing the lag between the disruption occurring and the plan responding to it.
- **Extend real-time information sharing upstream to the ingredient supplier**, in the same spirit as the module's CPFR content (synchronising inventory and planning strategies across firms) — earlier visibility of licensing timelines would let John's team see this risk before it forces a two-month postponement, rather than after.

Mark Allocation — Part (b) (worth 50% of the question)

Criterion	Marks	What earns the marks
Correctly outlines the standard S&OP process	12%	Identifies the cross-functional, cyclical nature of the process (baseline forecast → supply plan → agreed plan → implementation → review), applied to John's situation
Applies the process to address the immediate issue	13%	Explains specifically how/where in the cycle the 2-month delay should be raised and re-planned, referencing NPI/forecast-accuracy implications
Identifies specific, relevant improvements	15%	At least two distinct, well-justified improvements (e.g. earlier risk visibility, rolling/continuous planning, upstream information sharing)
Coherence and reasoning quality	10%	Answer connects the 'standard vs advanced' framing explicitly, rather than listing improvements without contrast
Total	50%	Sum of criteria above

Question 1 — total

100%

Question 2

40% + 60% = 100%

(a)**40%**

Identify and discuss the Supply Chain Design changes required to support a Demand Driven Supply Chain. Your answer should include the role of forecasting.

FULL MODEL ANSWER

The shift being described. A demand-driven supply chain is a *pull* system: it is triggered by actual customer demand signals rather than being pushed forward on the basis of a forecast made upstream. The module frames this explicitly as a response to the difficulty of forecasting and the costs of the bullwhip effect — moving supply chains from push (supply-driven) toward pull (demand-driven) wherever possible.

Design changes required:

- **Improve information sharing between firms — CPFR.** Collaborative Planning, Forecasting and Replenishment integrates multiple firms across the supply chain into a shared demand-planning process, so that inventory strategies are synchronised up and down the chain rather than each tier forecasting in isolation.
- **Access to real-time information.** This includes visibility of production schedules and mechanisms such as electronic Kanban, which support Vendor Managed Inventory (VMI) and/or Just-In-Time (JIT) replenishment — replacing periodic, forecast-triggered orders with signals tied to actual consumption.
- **Redesigning the trigger for replenishment** from a push schedule to a pull signal wherever the supply chain's lead times and product characteristics allow it.

The role of forecasting. A genuinely demand-driven design does not eliminate forecasting — the module is explicit that "forecasting may be required further upstream." Where lead times are long (raw materials, capacity planning, supplier scheduling), a forecast is still unavoidable, because a pull signal from the point of consumption cannot travel upstream faster than the product itself can be made and delivered. What changes is *where* forecasting sits: it moves away from the customer-facing end of the chain (where actual demand signals now trigger replenishment) and concentrates further upstream, closer to the constraints that genuinely require lead time.

This connects directly to the module's inventory reduction principles: pooling inventory, reducing demand variation, and reducing lead time all lower the safety stock required to protect against forecast error. Demand-driven design supports exactly these mechanisms — shorter, pull-triggered replenishment cycles reduce the distance (in time) over which a forecast has to hold, which is also the core logic behind why improved demand-stream visibility reduces bullwhip amplification and lag across the chain.

Mark Allocation — Part (a) (worth 40% of the question)

Criterion	Marks	What earns the marks
Correctly frames push vs pull / demand-driven logic	8%	Clearly explains why the shift responds to forecasting difficulty and bullwhip cost
Identifies CPFR / information-sharing design change	8%	Names and explains CPFR's role in synchronising inventory strategies across firms
Identifies real-time information / VMI / JIT design change	8%	Explains how real-time visibility supports pull-based replenishment
Role of forecasting explicitly addressed	10%	Explains that forecasting persists but shifts upstream to long-lead-time stages — not that it disappears
Links to bullwhip / inventory reduction principles	6%	Connects design changes to reduced safety stock / reduced bullwhip, showing integrated understanding
Total	40%	Sum of criteria above

(b)(1)**30%**

Sophie is a Supply Planner at a large global beverage manufacturing company. She has received an urgent message from the Sales team as follows: One of the company's largest supermarket customers has just placed an unexpectedly high order for a popular beverage due to a heatwave coinciding with a number of major music and sporting events in their region. The standard lead-time is 2 days. The order volume is double the normal monthly forecast for this Stock Keeping Unit (SKU). Sophie must act quickly to determine whether the business can accommodate this sudden demand without compromising supply for other key customers.

(1) Who will Sophie need to work with in the supply chain on this order?

FULL MODEL ANSWER

This is an immediate, cross-functional response problem, mirroring the cross-functional composition the module assigns to S&OP decision-making (sales, operations, purchasing, logistics, finance):

- **Production/Operations planning** — to check whether output can be flexed within the 2-day lead time, and what capacity trade-offs that requires against other scheduled runs.
- **Procurement** — to confirm raw material and packaging availability for double the forecast volume; a beverage SKU spike of this size may be constrained by packaging or ingredient stock, not just production capacity.
- **Logistics/Distribution** — to confirm transport and warehouse capacity can physically move double volume within the standard 2-day lead-time.
- **Warehouse/DC inventory teams** — to check current stock-on-hand for this SKU before assuming the gap must be closed entirely through incremental production.
- **Key Account/Customer Service teams for other major customers** — because the question explicitly frames this as a decision that must not compromise supply to other key accounts; this is an allocation decision, not just a fulfilment one, so whoever owns those other relationships needs to be in the loop before capacity is committed.
- **Sales** — to confirm the order is firm and to relay realistic delivery expectations back to the customer.

The rationale in each case is the same: a 2-day lead-time order at double the forecast cannot be resolved by one function acting alone — it requires the same kind of rapid, cross-functional alignment the module describes for S&OP decision-making, compressed into days rather than a monthly cycle.

Mark Allocation — Part (b)(1) (worth 30% of the question)

Criterion	Marks	What earns the marks
Identifies internal supply-side functions	12%	Production/operations, procurement, logistics/warehouse all identified with a specific reason for each
Identifies the allocation/other-customer dimension	10%	Explicitly names the need to involve teams responsible for other key accounts, not just fulfil this order in isolation
Rationale quality	8%	Explains why cross-functional, rapid coordination is required given the 2-day lead-time constraint
Total	30%	Sum of criteria above

(b)(2)**30%**

(Scenario as above — Sophie, Supply Planner, sudden double-forecast order, 2-day lead-time.)

(2) This order is twice the forecast for this SKU for this month. Who should Sophie collaborate with to avoid this situation in the future? Your answer should include the rationale informing your response.

FULL MODEL ANSWER

This second question is not about fulfilment — it is about forecast quality, so the collaboration required is different in kind from part (1):

- **The customer's own planning/category team, under a CPFR-style arrangement.**
The module defines CPFR as integrating multiple firms across the supply chain into the demand-planning process to synchronise inventory strategies. A heatwave coinciding with major regional events is exactly the kind of demand driver the retailer would likely see coming (via their own promotional and events calendar) before the manufacturer does — collaborating directly with the customer closes that visibility gap.
- **Internal Sales and Trade/Marketing teams** — who typically hold promotional calendars and market intelligence that should feed into the baseline forecast at the first stage of the S&OP cycle (gathering data from sales and marketing to build the demand plan). If a known event-driven demand spike wasn't captured there, that is the root cause, not a forecasting-model failure.
- **Demand planning, to build weather/event sensitivity into the forecasting method** — for example via a consensus panel that draws in qualitative, region-specific input (comparable to the module's Delphi/consensus panel approach) rather than relying solely on a historical-average quantitative forecast, which by construction cannot anticipate a one-off demand driver like this.

Rationale. The order was not really "unexpected" in the sense of being unforecastable — a heatwave and a cluster of major events are observable in advance. The gap is one of information flow, not inherent demand volatility: exactly the problem CPFR and a properly resourced S&OP baseline-forecast step are designed to close. Collaborating downstream (with the customer) and internally (sales/marketing feeding the forecast) addresses the cause; collaborating only with production or logistics, as in part (1), would only ever address the symptom.

Mark Allocation — Part (b)(2) (worth 30% of the question)

Criterion	Marks	What earns the marks
Identifies customer-facing/ CPFR-style collaboration	12%	Explains collaborating directly with the customer to access their promotional/events visibility
Identifies internal sales/ marketing forecast-input gap	10%	Links the miss to the S&OP baseline-forecast step, not just a production issue
Rationale distinguishes cause vs symptom	8%	Clearly explains why this is a forecasting/information-flow fix, distinct from part (1)'s fulfilment fix
Total	30%	Sum of criteria above

Question 2 — total**100%**

Question 3

60% + 40% = 100%

(a)**60%**

Using the example of grocery supply chains, illustrate and discuss the supply chain design before and after the introduction of central distribution, highlighting performance implications. Your answer should include basic before and after supply chain design schematics and identify key metric(s) when discussing performance.

FULL MODEL ANSWER

Before central distribution. Consumer food processors deliver directly to individual retail stores, typically limited to around twice-weekly delivery. Each store holds and manages its own inventory independently — there is no pooling of stock across the retailer's network, and delivery frequency is constrained by the economics of visiting many individual stores directly.

After central distribution. Delivery shifts to daily supplier deliveries into a central (or regional) distribution centre, with onward daily delivery to retail stores driven by daily store orders. Logistics arrangements range from delivery into the RDC to backhaul by the retailer's own distribution fleet from the supplier's factory gate (Factory Gate Pricing), and RDC operations frequently involve cross-docking — immediate despatch of supplier orders from the RDC to stores without being taken into stock — often run by an appointed 3PL.



Figure 1 — Grocery supply chain design before and after the introduction of central distribution. Arrows indicate physical product flow; inventory pools at store level before, and at the RDC after.

Performance implications:

- **Reduced days inventory on-hand throughout the chain** — pooling stock at the RDC rather than fragmenting it across every individual store lowers total inventory required for the same service level, and cross-docking further reduces RDC dwell time.
- **Improved information flow, reducing bullwhip** — daily, order-driven replenishment gives much tighter, higher-frequency demand signals back up the chain than twice-weekly direct delivery, reducing the amplification and lag that drive the bullwhip effect.
- **Enhanced stock turnover and on-shelf variety at retail** — more frequent, smaller, order-driven deliveries let stores carry a wider range without a proportional increase in backroom stock.

- **Requires continuous replenishment capability and high supplier compliance** — daily, order-driven delivery only works if suppliers reliably deliver on time and in full; this is why OTIF (On Time In Full — the "perfect order" concept) becomes a key operating metric under this model.
- **Opportunity for suppliers** — closer, higher-frequency integration with a large retailer creates the basis for a category-leadership relationship, though it also concentrates dependency on that retailer's central distribution requirements.

Key metrics. The module identifies inventory turnover, the cash-to-cash cycle time, and (retailer-specific) sales per m² as the core metrics for assessing the performance implications of this shift, alongside OTIF as the compliance metric that continuous replenishment depends on.

Mark Allocation — Part (a) (worth 60% of the question)

Criterion	Marks	What earns the marks
Accurately describes the 'before' design	10%	Direct-to-store delivery, twice-weekly, store-level inventory correctly identified
Accurately describes the 'after' design	10%	Daily RDC-based delivery, cross-docking, 3PL/backhaul correctly identified
Schematic diagrams provided (before and after)	10%	Basic before/after schematics included and broadly correct in structure
Performance implications discussed	15%	At least three implications discussed with correct reasoning (inventory reduction, bullwhip/information flow, turnover/variety, supplier compliance)
Key metric(s) correctly identified	10%	Inventory turnover AND cash-to-cash cycle AND/OR sales per m ² AND OTIF, correctly tied to the discussion
Overall coherence and use of grocery example	5%	Answer stays grounded in the grocery central distribution example throughout, not generic supply chain theory
Total	60%	Sum of criteria above

(b)**40%**

CW Foods is part of a large food multinational and supplies a range of ready meals. While this is a competitive market, they have differentiated their business from the competition through an exciting product range. This has positioned them as the main supplier to leading supermarkets across a number of European markets. Their ability to do this is supported by a wide range of high quality, speciality ingredient suppliers. These ingredients differentiate their ready meals from competitors. Many of these ingredient suppliers are small-scale businesses and are very responsive to orders, which minimizes ingredient inventory stocks on hand. As many of these suppliers are small-scale speciality businesses, CW Foods recognises the importance of cashflow. A policy for all suppliers of payment at 30 days after receipt of goods has been in place for many years. At Group level, Finance have recently introduced a Group-wide policy of 50 days for accounts payable. Finance at Corporate HQ have been reviewing cash-to-cash cycles in all the Group's businesses. The CW Foods review reports a relatively low inventory (an average of 35 days inventory on-hand) and accounts receivable of 60 days conforms with corporate policy. However, as part of their review, Finance have asked CW Foods to increase their accounts payable to 50 days. Calculate the cash-to-cash cycle time both currently and what it would be, given the implementation of the corporate accounts payable policy. How would you respond to the request from Finance at HQ?

FULL MODEL ANSWER

Cash-to-cash cycle formula (module content): Days Inventory On-hand + Days Sales Outstanding – Days Payables Outstanding = Cash-to-cash days.

Scenario	DIO	DSO	DPO	Cash-to-cash
Current (30-day supplier terms)	35	60	30	65 days
Proposed (Group 50-day policy)	35	60	50	45 days

$35 + 60 - 30 = \mathbf{65 \text{ days}}$ currently. $35 + 60 - 50 = \mathbf{45 \text{ days}}$ under the Group's proposed policy — a **20-day reduction** in the cash-to-cash cycle, which releases cash and is favourable purely from a Group cash-conversion perspective.

Verdict — Numerically favourable to Finance, but should not be implemented as a blanket policy for CW Foods

How to respond. The calculation is straightforward, but implementing it uniformly ignores exactly the trade-off the module identifies: Finance's objective (conserve funds, improve the Group's reported cash position) is not automatically aligned with Supply Chain's objective (continuity and quality of supply at a fair price). CW Foods' competitive differentiation depends specifically on speciality ingredient suppliers who are described as small-scale and who already minimise their own inventory to stay responsive. That combination — small scale, low buffer stock, high responsiveness — means these suppliers are precisely the ones

most exposed to a cashflow shock from extending payment terms from 30 to 50 days; CW Foods' own policy rationale for the original 30-day terms was explicitly the importance of these suppliers' cashflow.

The appropriate response to Finance is therefore not a flat refusal or a silent roll-out, but a differentiated implementation, informed by supplier and customer profiling as the module recommends:

- **Segment suppliers before applying the new terms.** Apply the 50-day policy to larger, financially robust suppliers where it carries little risk, while protecting the small-scale speciality suppliers who are core to the differentiated product range and least able to absorb a 20-day extension.
- **Offer supply chain finance / early-payment facilities** for the suppliers where extended terms would otherwise threaten continuity — the supplier is paid early via a financing arrangement while CW Foods still books the 50-day term, preserving the Group's cash-to-cash improvement without pushing cashflow risk onto suppliers CW Foods cannot afford to lose.
- **Feed this trade-off back to Corporate HQ explicitly** rather than applying the policy silently: Finance may be several organisational levels removed from the supplier relationships underpinning CW Foods' market differentiation, and needs that context to assess whether a blanket 50-day policy is actually the right instrument here versus a segmented approach.

Mark Allocation — Part (b) (worth 40% of the question)

Criterion	Marks	What earns the marks
Correct formula applied	5%	DIO + DSO – DPO stated and applied correctly
Correct current cash-to-cash calculation	5%	$35 + 60 - 30 = 65$ days, shown with full working
Correct proposed cash-to-cash calculation	5%	$35 + 60 - 50 = 45$ days, shown with full working, and the 20-day change correctly identified
Identifies the Finance vs Supply Chain trade-off	10%	Recognises that a numerically favourable change carries supplier-continuity risk for CW Foods specifically
Response is specific to CW Foods' supplier base	10%	Answer explicitly connects the risk to the small-scale, low-inventory speciality suppliers described in the scenario, not a generic answer
Proposes a workable response	5%	At least one concrete mechanism (segmentation, supply chain finance, escalation with rationale to HQ)
Total	40%	Sum of criteria above

Question 3 — total

100%